

Where the Gigabytes Went · AS 9618 — ANSWERS (student copy) · first edition, the sheet set as homework

1. Draw one line — 1 mark per correct line.

Left	Right	because
1 byte	8 bits	the definition
1 kilobyte	8 000 bits	1000 bytes × 8
1 kibibyte	8 192 bits	1024 bytes × 8
1 megabyte	1 000 kilobytes	decimal ladder, ×1000
1 tebibyte	1 024 gibibytes	binary ladder, ×1024

Unused: 1 024 kilobytes — which is 1 024 000 bytes and equals *nothing* on the left. That spare is the whole question. It catches the student who has learned "1024 of the previous one" as a spell and applies it to a *decimal* name. 1024 kilobytes is not a mebibyte (1 048 576 bytes) and not anything else with a name — it is a number somebody made by mixing the ladders. Say it out loud: **1024 goes with kibi, 1000 goes with kilo, and they never cross.**

2. Prefixes — 1 mark per blank in (a), 1 for (b).

(a) A mebibyte has a **binary** prefix. Four mebibytes = **4096** kibibytes. A gigabyte has a **decimal / denary** prefix. Three terabytes = **3000** gigabytes.

(b) **2 mebibytes.**

2000 KiB = 2 048 000 2 MiB = 2 097 152 ← largest
0.2 MB = 200 000 2000 kB = 2 000 000

Accept "denary" for decimal — it is what the rest of the course says. **Three of the four are within 5 % of each other and the fourth is a tenth the size** — 0.2 megabytes is there purely to catch skimming.

3. A bitmap image — 2 marks for working, 1 for the answer; 1 for (b).

1024 × 768 = 786 432 pixels
786 432 × 24 = 18 874 368 bits
18 874 368 ÷ 8 = 2 359 296 bytes
2 359 296 ÷ 1024 ÷ 1024 = 2.25 MiB

(b) The file also stores a **file header** — the width, the height, the colour depth and which format it is — before the first pixel. (*Accept: a colour palette / other metadata.*) This is the one place 9618's wording differs from 0478's, and it is deliberate: 9618 says **estimate**, 0478 says **calculate**. Both mean the same arithmetic; only one of them is being honest about the header. Do not accept "because of compression" — a compressed file is *smaller*, not larger, and the question asked why the real file is bigger.

4. A sound file — 2 marks for working, 1 for the answer; 1 for (b).

16 384 × 16 × 32 = 8 388 608 bits
8 388 608 ÷ 8 = 1 048 576 bytes
1 048 576 ÷ 1024 = 1024 KiB

(b) Halve the rate, halve the samples, halve the file: **512 KiB.**

 **Unhinted korok** — the answer is exactly **1 MiB**. 1024 KiB is one mebibyte on the nose, and it is no accident: $16\,384 = 2^{14}$, $16 = 2^4$, $32 = 2^5$, so the whole calculation is $2^{14} \times 2^4 \times 2^5 = 2^{23}$ bits = 2^{20} bytes = 1 MiB. **Every single number in the question is a power of two.** The question asks for KiB precisely so that the 1024 sits there looking innocent. Nothing on the sheet points at it.

5. Find the mistake — 1 for naming it, 2 for the corrected working and answer.

The student divided **bits** by 1024. Bits must be converted to **bytes** first — the ladder above the byte is a ladder of *bytes*.

3 840 000 ÷ 8 = 480 000 bytes
480 000 ÷ 1024 = 468.75 KiB

Their number is exactly **8 times too big**, which is the tell. The check: if an answer in KiB is larger than the pixel count, something has gone wrong, because 8-bit colour is one byte per pixel and there are 1024 bytes in a KiB. Accept 468.75 or 469 KiB (rounding stated).

6. Gibibyte vs gigabyte — 1 mark, and the answer needs *both* numbers.

One gibibyte is **1024 mebibytes** ($2^{30} = 1\,073\,741\,824$ bytes); one gigabyte is **1000 megabytes** (10^9 bytes). Reject "a gibibyte is bigger" on its own — true, and worth nothing. The mark scheme wants the two ladders named or numbered. This is the exact wording of a real 9618 one-marker, asked about tebi/tera in 2022 and kibi/kilo in 2023, so it will come round again.

Total: 22 marks.

 **The hinted** — (the top of the ladder): zebi (2^{70}) and yobi (2^{80}) are real, standard, and were long treated as absurd. Then storage caught up. In **November 2022** the General Conference on Weights and Measures added **ronna** (10^{27}) and **quetta** (10^{30}) to the metric system — the first new prefixes since 1991, and the stated reason was that the data world was about to run out of ladder. The number a curious student will find is that global data production is now measured in the low hundreds of **zettabytes** a year, which puts us one rung below yotta and climbing. Somebody had to add two more rungs to a system that had not needed one in thirty years, because of files.